

RAVI RAJ

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Summary

Aspiring Software Engineer with a strong foundation in Object-Oriented Programming (OOP), Data Structures & Algorithms, Database Management Systems (DBMS), Operating Systems, and Software Development Lifecycle (SDLC). Hands-on experience in designing, developing, testing, maintaining, and optimizing software applications using Python, SQL, REST APIs, and relational databases. Familiar with Agile methodologies, version control using Git, debugging, and unit testing fundamentals. Passionate about writing clean, maintainable code and solving real-world engineering problems through analytical thinking and collaborative software development.

Technical Skills

Programming Languages: Python, C++

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems

Backend Technologies: Flask, REST APIs, FAST APIs

Database: MySQL, SQL Server, SQLite

Web Technologies: HTML, CSS, JavaScript (Basic)

Developer Tools: Git, GitHub, Linux, VS Code

Software Engineering: Agile Methodology, SDLC, Version Control (Git), REST APIs, Code Debugging, Problem Solving

Key Projects

Inventory Management & Database Pipeline

[View](#)

Python | SQL | SQLite | MySQL | Git | GitHub

- Designed a 5-table relational database schema using foreign keys to ensure data integrity across inventory records.
- Developed Python CRUD operations and optimized SQL queries for efficient stock tracking and inventory management.
- Implemented low-stock alerts, modular CLI workflows, and managed source code using Git and GitHub.

Autonomous DNN Repair Using Gradient Detection and LoRA

[View](#)

Python | NumPy | SciPy

- Developed an automated framework for detecting and repairing neural network faults using LoRA fine-tuning.
- Improved execution speed by 15× through algorithm optimization while enhancing model reliability.
- Applied debugging, testing, and performance optimization to improve software reliability and maintainability.

Handwritten Devanagari Character Recognition Using SCNN

[View](#)

Python | OpenCV | Flask | SQL Server

- Developed an SCNN-based handwritten character recognition system achieving 93.91% accuracy on over 92,000 images.
- Built image preprocessing pipelines and Flask REST APIs for real-time handwritten character prediction.
- Optimized database operations through debugging, testing, and code optimization to improve application performance.

Achievements

CSBIZ Nexus Society

2024–2025

- Assisted the management team in society operations, prepared meeting minutes, coordinated workshops with industry professionals, and collaborated with teams to ensure successful technical events.

Education

Chandigarh University, Mohali

2022 – 2026

Bachelor of Engineering in Computer Science & Business Systems

CGPA: 7.94

Patel College, Bihar

2021

Intermediate (BSEB)

First Div

P.P. Public School, Bihar

2019

High School (CBSE)

First div

Certifications

Oracle Cloud Infrastructure – Data Science

[View](#)

Generative AI Powered Data Analytics – Tata (Forage)

[View](#)

Key Strengths

Problem Solving • Data Structures & Algorithms • Object-Oriented Programming • Database Design • Software Development • REST API Development • Debugging & Testing • Version Control (Git) • Team Collaboration • Quick Learner